Agent-Based Simulation of Organisational Misconduct

Reporting, Enforcement, and Hidden Misconduct

Presentation of Group 5

Daniel Steiner, Joel Stehlin, Marco Birchler and Raphael Bosshart

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1. Motivation and Research Question

43%

of occupational fraud cases are detected through tips

ACFE, 2024

46%

of reported cases are confirmed as real misconduct after investigation

NAVEX, 2025

54%

of people who reported wrongdoing said they felt pressure not to report

EY Global Integrity Report, 2024

Research Question

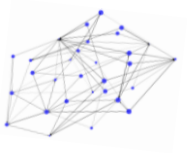
Under what conditions does stricter enforcement reduce true misconduct

We are interested in the gap between:

Visible / Punished misconduct and total misconduct in the organisation.

Why ABM

Phenomenon depends on **agents with heterogeneous attributes, decentralised behaviour, dynamic dependencies, and emergent macro outcomes**



2. The Central Paradigm & Non-Linearity

Why Non-Linear

- **Enforcement** affects both **misconduct** and **reporting**
- **Reporting** depends on fear, and protection
- **Misconduct** becomes visible only through observation and reporting

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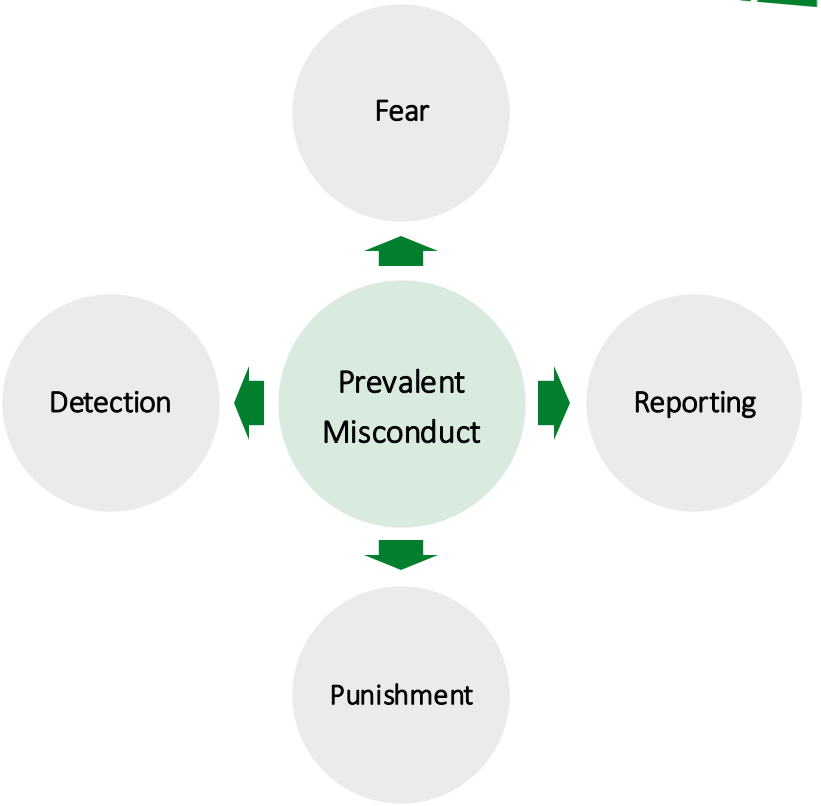
Stricter enforcement can lead to increased retaliation which in turn reduces reported misconduct while hidden misconduct increases

Zero tolerance might only be optimal if the organization preserves a save reporting climate and keeps fear of retaliation at low levels



3. Agents and Environment

Agents

One Agent Type:
Employees

They can both

- Commit misconduct
- Report misconduct

Environment

One room with a fixed population of employees

- Agents move around and interact through observation of misconduct
- Enforcement and sanctions are determined by global policy parameters
- Propensity of misconduct and fear of retaliation evolve over time

(Reductionistic)

Deliberately kept only the essential structure needed to model misconduct, reporting, and feedback

4. Agent Attributes

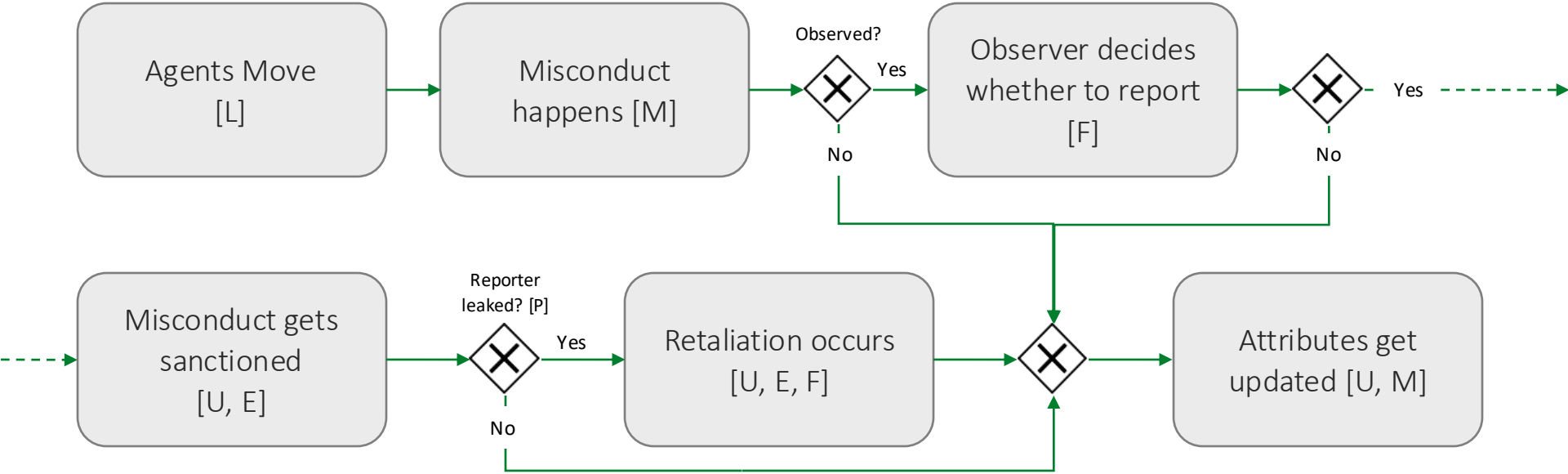
Attribute		Meaning
Employee Utility	U	Personal Utility attributed to each Employee. Increases by successful misconduct. Decreases by Punishment & Retaliation
Misconduct propensity	M	Likelihood that an employee engages in misconduct for personal benefit at any point in time
Fear of retaliation	F	Perceived personal risk associated with reporting other agents for wrongdoing
Location	L	Current location of the Agent, relevant for observing misconduct

5. Environmental Parameters

Parameter		Meaning
Enforcement strictness	E	Severity of punishment for misconduct
Reporter protection	P	How well the organization protects reporting employees from retaliation

6. What happens per Tick

Each simulation tick updates misconduct, reporting, investigation, and feedback.



7. Key Outputs

Research Question

Under what conditions does stricter enforcement reduce true misconduct

Output	Meaning
True misconduct	Total amount of misconduct actually occurring in the organization
Hidden misconduct	Misconduct that occurs but is not sanctioned or detected
Hidden Misconduct Rate	= (True Misconduct - Sanctioned Misconduct) / True Misconduct. **Captures the key question**: is the organization reducing misconduct itself, or only reducing what becomes visible?

→ **Vary policy conditions** (Ex. weak, strong enforcement) and compare resulting outputs.

Thank you

Questions and feedback

Appendix: What is Left Out & Future Iterative Extensions

What is Left Out

- Detailed employee characteristics
  - (Gender, Work Hours, Quality of work, length of employment, nature of role)

Future Iterative Extensions

- Separate Investigation of Misconduct
  - Evidence Quality
- Warning Rate / # Warnings before Punishment
- Compliance unit
- Managers and compliance officers as new agents
- Seniority Agent Attribute